CURRICULUM VITAE

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Citizenship:	Canada

EDUCATION

Sept. 1985– Aug. 1989	**Dept. of Electrical Engineering and Computer Science** and **Operations Research Center**, **Massachusetts Institute of Technology**, Cambridge, Mass., USA - Candidate for Ph.D; completed August 1989 - Minor in Computer Science; - Thesis topic: *Communication Complexity of Some Problems in Distributed Computation*; (Supervisor: J.N. Tsitsiklis)
Sept. 1984– May 1985	**Nankai Institute of Mathematics, Nankai University**, Tianjin, China - One year intensive training program in Mathematics and English, sponsored by the Ministry of Education of China.

Sept. 1980– May 1984	**Dept. of Mathematics, Peking University**, Beijing, China - B.Sc. in Applied Mathematics, 1984; GPA 4.9/5 - Awarded several scholarships for outstanding academic performances, 1980-84. - Selected as one of the 15 students nationally by the Government of China and by a joint SIAM-AMS (*Society of Industrial and Applied Mathematics* and *American Mathematical Society*) committee for Ph.D study in the U.S. with full scholarship, 1984. (Also known as the S.S. Chern Program)

EMPLOYMENT HISTORY

Sept. 2025– present	**Shenzhen Loop Area Institute, Shenzhen, China** *Executive Director*
March 2015– present	**Shenzhen Research Institute of Big Data, Shenzhen, China** *Director*
July 2014– present	**The Chinese University of Hong Kong, Shenzhen** *Vice President (Academic), Presidential Chair Professor*
April 2003– present	**Dept. of Electrical and Computer Engineering, University of Minnesota** *Professor*, on leave since 2014
April 2003– July 2013	**Dept. of Electrical and Computer Engineering, University of Minnesota** *Professor and ADC Chair in Digital Technology*
July 2001– July 2004	**Dept. of Electrical and Computer Engineering, McMaster University** *Canada Research Chair (Tier I) in Information Processing*
July 2000– April 2003	**Dept. of Electrical and Computer Engineering, McMaster University** *Department Head*
July 1998– June 2005	**Dept. of Electrical and Computer Engineering, McMaster University** *Full Professor*
July 1992– June 2003	**Dept. of Computing and Software, McMaster University** *Associate member*
July 1993– June 1998	**Dept. of Electrical and Computer Engineering, McMaster University** *Tenured Associate Professor*
Sept. 1989– June 1993	**Dept. of Electrical and Computer Engineering, McMaster University** *Tenure Track Assistant Professor*

Jan. 1986– Aug. 1989	**Laboratory for Information and Decision Systems, MIT** *Research Assistant* Research to determine the minimum amount of information that has to be exchanged in order for a set of processors to solve a problem cooperatively.
Sept. 1985– Dec. 1985	**Center for Technology, Policy and Industrial Development, MIT** *Research Assistant*

AWARDS

1. 2025 First Hua Prize from International Congress of Chinese Mathematicians.
2. 2023 Shenzhen Science and Technology Progress Award, First Place.
3. 2022 First CSIAM Wang Xuan Applied Mathematics Prize.
4. Foreign member of the Chinese Academy of Engineering (CAE), 2021.
5. 2020 ICCM Best Paper Award from the International Consortium of Chinese Mathematicians for the paper: Mingyi Hong, Zhi-Quan Luo, and Meisam Razaviyayn, "Convergence Analysis of Alternating Direction Method of Multipliers for a Family of Nonconvex Problems", *SIAM Journal on Optimization*, Vol. 26, No. 1, Jan. 2016.
6. 2018 Paul Y. Tseng Memorial Lectureship in Continuous Optimization from Mathematical Optimization Society, for outstanding contributions in continuous optimization.
7. 2015 IEEE Signal Processing Society Best Magazine Paper Award for the paper: Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye and S. Zhang, "Semidefinite Relaxation Of Quadratic Optimization Problems", *IEEE Signal Processing Magazine*, Vol. 27, No. 3, May 2010.
8. Fellow of the Royal Society of Canada, 2014, for his influential contributions to optimization algorithms and their applications in signal processing, communications.
9. 2011 IEEE Signal Processing Society Best Paper Award for the paper: Sidiropoulos, N.D., Davidson, T.N., and Luo, Z.-Q., "Transmit Beamforming for Physical Layer Multicasting," *IEEE Transactions on Signal Processing*, Vol. 54, No. 6, pp. 2239–2251, 2006.
10. Best Paper Award, *2011 IEEE International Conference on Communications*, for the paper: Y.-F. Liu, Y.-H. Dai, and Z.-Q. Luo, "Max-Min Fairness Linear Transceiver Design for a Multi-User MIMO Interference Channel".
11. SIAM Fellow, 2011, for the development of novel applied mathematics ideas and methods for signal processing and digital communication.
12. 2011 EURASIP Best Paper Award, for the paper: Luo, Z.-Q. and Pang, J.-S., "Analysis of Iterative Waterfilling Algorithm for Multiuser Power Control in Digital Subscriber Lines", published in 2006 in the *EURASIP Journal on Applied Signal Processing*.
13. 2010 Farkas Prize from the Institute for Operations Research and the Management Sciences, for outstanding contributions to the field of optimization.
14. 2009 IEEE Signal Processing Society Best Paper Award for the paper: Luo, Z.-Q. and Zhang, S., "Dynamic Spectrum Management: Complexity and Duality," *IEEE Journal of Selected Topics in Signal Processing*, Special Issue on Signal Processing and Networking for Dynamic Spectrum Access, Vol. 2, No. 1, pp. 57–73, February 2008.
15. Fellow of IEEE, 2007.

16. 2004 ICASSP Best Student Paper Award (with Ph.D student M. Kisialiou).
17. 2004 IEEE Signal Processing Society Best Paper Award for the paper "Robust Adaptive Beamforming Using Worst-Case Performance Optimization: A Solution to the Signal Mismatch Problem" which appeared in *IEEE Transactions on Signal Processing*, (co-authors: Vorobyov and Gershman), February 2003.
18. ADC Chair in Digital Technology, University of Minnesota, April 2003 – July 2013.
19. Canada Research Chair (Tier I) in Information Processing, McMaster University, July 2001-July 2004.
20. Best Paper Award in the *International Conference on Optimization Techniques and Applications*, for the paper "Optimal Transceiver Design Via Convex Optimization", December 2001, Hong Kong.
21. International Research Fellowship from K.C. Wong Education Foundation (Hong Kong) through Chinese Academy of Science, 1996.

PROFESSIONAL ACTIVITIES

1. Member of The Shaw Prize Council, 2025-2026
2. Technical Program Committee Chair, ICCOPT 2025.
3. Member of IEEE SPS Awards Board, 2024-2026.
4. Chair, Selection Committee for the Paul Y. Tseng Memorial Lectureship in Continuous Optimization, Mathematical Optimization Society, 2021.
5. Member of IEEE Alexander Graham Bell Medal Selection Committee, 2018-2020.
6. Member of IEEE SPS Awards Board, 2017-2018.
7. Member of IEEE SPS Fellow Selection Committee, 2017-2019.
8. Chair, Farkas Prize Selection Committee, INFORMS Optimization Society, 2015.
9. General co-chair, ICASSP 2016, Shanghai, China.
10. Editor in Chief, IEEE Transactions on Signal Processing, 2012-2014.
11. Visiting Professor, EECS Department, MIT, Spring 2011.
12. Past Chair (2012-2013)/Chair (2011-2012)/Vice chair (2009-2010), IEEE Signal Processing Society Technical Committee on Signal Processing for Communications (SPCOM).
13. Member of IEEE Signal Processing Society Technical Committees on Signal Processing for Communications (SPCOM), 2006-2012, and on Signal Processing Theory and Methods (SPTM), 2006-2009.

14. Distinguished Lecturer for the *IEEE Sensor Array and Multichannel Signal Processing Workshop*, Waltham, MA, June 2006.

15. Semi-plenary speaker, *International Symposium on Mathematical Programming*, Copenhagen, Denmark, August, 2003.

16. Invited tutorial on "Convex Optimization Techniques for Signal Processing", *IEEE International Conference on Acoustics Speech and Signal Processing,* Orlando, Florida, 2002.

17. Visiting Professor (September 2000 – December 2000), Department of Applied Mathematics and Physics, School of Informatics, Kyoto University, Kyoto, Japan.

18. Visiting Professor (September 2003 – December 2003), Information Systems Laboratory, Department of Electrical Engineering, Stanford University, USA.

19. Visiting Examiner (2003 – 2006), Department of Systems Engineering and Engineering Management, Chinese University of Hong Kong, Hong Kong.

20. International Research Fellowship from K.C. Wong Education Foundation (Hong Kong) through Chinese Academy of Science, 1996.

21. Guest co-editor, special issue of *IEEE Journal of Selected Areas of Communications* on 'Nonlinear Optimization of Communication Systems', 2006.

22. Guest co-editor for a special issue of *IEEE Signal Processing Magazine* on Distributed Signal Processing for Sensor Networks, 2006.

23. Guest co-editor, special issue of *Eurasip Journal on Applied Signal Processing*, on 'Multisensor Processing for Signal Extraction and Applications', 2006.

24. Associate editor for

    - *Management Science*, since 2009
    - *Journal of Optimization Theory and Applications*, 1991–2003
    - *SIAM Journal on Optimization*, 1998-2003
    - *Mathematics of Computation*, 1999-2004
    - *Mathematical Programming*, series B, 2003-2008
    - *Mathematics of Operations Research*, 2010-2015
    - *IEEE Transactions on Signal Processing*, 2000–2004
    - *Optimization and Engineering*, 1996-1999
    - *IEEE Signal Processing Magazine*, 2010-2014

25. Guest Professor, School of Mathematical Sciences, Peking University, P.R. China, 2009–2011.

26. Adjunct Professor, Department of Information Science and Engineering, Nanchang University, Jiangxi, P.R. China.

27. Adjunct Professor, Department of Applied Mathematics, Shanghai University, Shanghai, P.R. China.

28. Referee of numerous NSF (U.S.), NSERC (Canada), and RGC (Hong Kong) grant applications on Applied Mathematics, Signal Processing, Circuits and Systems

29. Served on CITO (*Communication and Information Technology Ontario*, a provincial research center of excellence) research program review panel and NSF proposal review panel in the area of computational sciences and information technology.

30. Organizer of 1998 international workshop on Optimization Methods and Applications, held in City University of Hong Kong.

31. Organizer and session chairperson of many international conferences on high performance computing and optimization methods.

32. Nominated twice for the McMaster University teaching award by the McMaster Student Union.

RESEARCH FUNDING (1997 – present)

1. Guangdong Province Department of Science and Technology, RMB 20,000,000, December 2023 – December 2028
   - Project title: Guangdong Major Project of Basic and Applied Basic Research
   - Principal investigator: Luo, Z.-Q.
2. National Natural Science Foundation of China, RMB 2,800,000, January 2018 – December 2022.
   - Project title: Big Data Analysis Technology and Application Based on Distributed Non-convex Statistical Model
   - Principal investigator: Luo, Z.-Q.
3. National Natural Science Foundation of China, RMB 770,000, January 2016 – December 2019.
   - Project title: Joint Optimization Management of Ultra-dense Radio Access Network and Backhaul Network
   - Principal investigator: Luo, Z.-Q.
4. Shenzhen Development and Reform Commission, RMB 10,000,000, 2017-2020.
   - Project title: Discipline Construction of Data Science and Big Data Technology
   - Principal investigator: Luo, Z.-Q.
5. Huawei, RMB 17,970,000, August 2020 – August 2023.
   - Project title: Future Network Optimization Innovation Laboratory of Huawei, CUHK(SZ) and Shenzhen Research Institute of Big Data
   - Principal investigator: Luo, Z.-Q.
6. Tencent, RMB 4,000,000, June 2018 – August 2023.
   - Project title: Joint Laboratory of Machine Intelligence between CUHK(SZ)-Tencent AI Lab
   - Principal investigator: Luo, Z.-Q.
7. Guangdong Province Leading Talents Introduction Program, RMB 5,000,000, December 2015 – December 2020.
   - Project title: Massive Data Information Processing
   - Principal investigator: Luo, Z.-Q.
8. Huawei, RMB 1,970,000, October 2018 – October 2020.
   - Project title: Intelligent Collaborative Research on Distributed Network Resources
   - Principal investigator: Luo, Z.-Q.
9. Huawei, RMB 2,400,000, November 2018 – January 2020.

- Project title: Data-driven Frontier Application Technology Research Cooperation Project
- Principal investigator: Luo, Z.-Q.

10. Shenzhen Science and Technology Innovation Committee, RMB 30,000,000, November 2015 - October 2020.

    - Project title: Big Data Information Processing and Application Innovation Team
    - Principal investigator: Luo, Z.-Q.

11. National Science Foundation, $310,000, September 1 , 2015 – August 31, 2018.

    - Project title: Optimal Provision of Backhaul and Radio Access Networks: A Cross-Network Approach
    - Principal investigator: Luo, Z.-Q.

12. Huawei, $460,000, 01/01/2014-12/31/2017

    - Project title: Interference Management Techniques for Wireless Communication
    - Principal investigator: Luo, Z.-Q.

13. National Science Foundation, $314,125, August 1 , 2012 – July 31, 2015.

    - Project title: A Cross-Tier Approach to Interference Management in Wireless Heterogeneous Networks
    - Principal investigator: Luo, Z.-Q.

14. US AFOSR, $303,950, July 5, 2012– June 14, 2015.

    - Project title: Mixed Integer Quadratic Optimization
    - Principal investigator: Luo, Z.-Q.

15. Huawei, $410,000, 01/01/2010-12/31/2013

    - Project title: Interference Management Techniques for Wireless Communication
    - Principal investigator: Luo, Z.-Q.

16. NIH, $275,000, 01/01/2011-12/30/2012

    - Project title: Novel, Fast B1 Mapping with Low SAR for Multiple RF Coils at Ultra High Field
    - Principal investigator: P.F. Van de Moortele, Co-PI: Luo, Z.-Q.

17. US NSF, $120,000, 2010–2013.

    - Project title: Semidefinite Programming Relaxation: Approximation Algorithms, Performance Analysis and Applications
    - Principal investigator: Luo, Z.-Q.

18. US AFOSR, $280,949, 2009–2012.
    - Project title: Optimization Algorithms and Equilibrium Analysis for Dynamic Resource Allocation
    - Principal investigator: Luo, Z.-Q.
19. US DOD Army, $255,000, 2009–2012.
    - Project title: Dynamic Spectrum Management: Complexity, Duality and Approximation
    - Principal investigator: Luo, Z.-Q.
20. ADC Endowed Chair operating fund, $45,000 per annum, 2009-2014, University of Minnesota Foundation
    - Principal investigator: Luo, Z.-Q.
21. Digital Technology Center, University of Minnesota, $28,500, 2007-2008
    - Project title: Optimal B1 Shimming for Ultra High Field Human Scanners
    - Principal investigator: Luo, Z.-Q., Co-PI: Van de Moortele, P.F.
22. Institute of Technology, University of Minnesota, $10,000, 2007-2008
    - Project title: Mathematical Algorithms for Image Processing
    - Principal investigator: Luo, Z.-Q., Co-PI: Ming, Jiang (Peking University)
23. National Science Foundation (CMMI), $314,604, 2007–2010.
    - Project title: Optimal Resource Management: Complexity, Duality and Approximation
    - Principal investigator: Luo, Z.-Q.
24. National Science Foundation (DMS), $150,000, 2006–2009.
    - Project title: High Performance Approximation Algorithms for Nonconvex Quadratic Optimization with Applications in Signal Processing and Communication
    - Principal investigator: Luo, Z.-Q.
25. US DOD ARMY, $300,000, 2005–2008.
    - Project title: Mathematical Analysis of the Signal Processing Capabilities of Sensor Networks
    - Principal investigator: Luo, Z.-Q.
26. National Science Foundation (DMS), $188,000, 2003–2007.
    - Project title: Convex Optimization Techniques in Signal Processing and Communication
    - Principal investigator: Luo, Z.-Q.
27. ADC Chair operating fund, $50,000 per annum, 2003-2008, University of Minnesota Foundation

- Principal investigator: Luo, Z.-Q.

28. Operating grant from Canada Research Chairs program, $200,000 per annum, 2001–2008.
    - Principal investigator: Luo, Z.-Q.
29. Research grant from Ontario Innovation Trust, $49,660 per annum, 2001-2005.
    - Principal investigator: Luo, Z.-Q.
30. Research Grant from Canada Foundation for Innovation, $49,660 per annum, 2001-2005.
    - Project title: Testing and Analysis Equipment for Information Processing Lab.
    - Principal investigator: Luo, Z.-Q.
31. Research grant from NSERC, $42,000 per annum, 1999–2003.
    - Project title: Advanced Optimization Algorithms for Digital Communication and Signal Processing.
    - Principal investigator: Luo, Z.-Q.
32. Research contract from CITO, $90,000 per annum 2000-2002.
    - Project title: Data Fusion with Applications to Multi-target Tracking
    - Principal investigator: Luo, Z.-Q., Co-investigators: Bosse, E. and Wong, M.
33. Research contract from CITO, $75,000 per annum 2001-2003.
    - Project title: Algorithmic Issues in Practical Communication
    - Principal investigator: Luo, Z.-Q., Co-investigators: Davidson, T.N.
34. International Cooperative Research Project sponsored by the Ministry of Education, Science, Sports and Culture of Japan, $9,900,000 (Yen), 1999-2001.
    - Project title: Synthesis, Analysis, and Algorithms for Equilibrium Problems
    - Principal investigator: Fukushima, M. (Kyoto University, Japan), Co-investigators: Kanzow, C. (Germany), Luo, Z.-Q. (Canada), Qi, L. (Australia), Pang, J.S., Tseng, P. (U.S.A.), Chen, X., Yamashita (Japan).
35. Research contract from CITO, $50,000 per annum, 1998-2000.
    - Project title: Data Fusion with Applications to Multi-target Tracking
    - Principal investigator: Luo, Z.-Q., Co-investigators: Bosse, E. and Wong, M.
36. Research contract from Groupe Informission, $30,000 per annum, 1998-2000.
    - Project title: Data Fusion with Applications to Multi-target Tracking
    - Principal investigator: Luo, Z.-Q., Co-investigators: Bosse, E. and Wong, M.
37. Research contract from Nortel Networks, $22,000, 1998-1999.

- Project title: High Order Statistics Methods for Blind Signal Separation
- Principal Investigator: Luo, Z.-Q.

38. Research contract from Ontario Hydro Technology, $17,000, 1998-1999.

    - Project title: Feasibility Study of Digital Communication over Power Lines.
    - Principal investigator: Luo, Z.-Q., Co-investigators: Davidson, T., Reilly, J.P.

39. Research contract from DREV, $120,000, 1997-1999.

    - Project title: Multi-target Tracking and Data Fusion
    - Principal investigator: Luo, Z.-Q., Co-investigator: Wong, K.M.

40. Research grant from NSERC, $33,300 per annum, 1995–1999.

    - Project title: Mathematical Programs with Equilibrium Constraints
    - Principal investigator: Luo, Z.-Q.

41. Research contract from DREO, $74,405, 1997-1998.

    - Project title: DSP implementation of On-line Classification Algorithms for Intra-Pulse Analysis.
    - Principal investigator: Luo, Z.-Q., Co-investigator: Wong, K.M.

42. Research contract from DREO, $55,000, 1996-1997.

    - Project title: Intra-Pulse Analysis of Passive Emitters
    - Principal investigator: Wong, K.M., Co-investigator: Luo, Z.-Q.

SELECTED PUBLICATIONS

A. Refereed Journal Publications

- **Journal Papers (published or accepted)**

1. K. Li, W. Pu and Z.-Q. Luo, "Efficient Beamforming Refinement for Limited Feedback FDD Massive MIMO: An Online Alternating Exploration-Estimation Approach," *IEEE Transactions on Signal Processing*, Vol. 73, pp. 4969–4985, 2025.
2. W.-K. Chen, Y.-F. Liu, Y.-H. Dai and Z.-Q. Luo, "QoS-Aware and Routing-Flexible Network Slicing for Service-Oriented Networks," *IEEE Transactions on Network and Service Management*, Vol. 22, No. 6, pp. 6021–6036, Dec. 2025.
3. A. Maatouk, F. Ayed, N. Piovesan, A. D. Domenico, M. Debbah and Z.-Q. Luo, "TeleQnA: A Benchmark Dataset to Assess Large Language Models Telecommunications Knowledge," *IEEE Network*, 2025.
4. D. Rybin, Y. Zhang and Z.-Q. Luo, "XX^T Can Be Faster," *arXiv preprint arXiv:2505.09814*, 2025.
5. Y. Zhang, C. Chen, Z. Li, T. Ding, C. Wu, D. P. Kingma, Y. Ye, Z.-Q. Luo and R. Sun, "Adam-mini: Use Fewer Learning Rates to Gain More," *arXiv preprint arXiv:2406.16793*, 2024.
6. Y. Jiao, Y. Gu, T.-H. Chang and Z.-Q. Luo, "Decentralized Rank-Adaptive Matrix Factorization Part I: Algorithm Development," *IEEE Transactions on Signal Processing*, 2024.
7. W.-K. Chen, Z. Wu, R.-J. Zhang, Y.-F. Liu, Y.-H. Dai and Z.-Q. Luo, "An Efficient Benders Decomposition Approach for Optimal Large-Scale Network Slicing," *IEEE Transactions on Signal Processing*, Vol. 72, pp. 4935–4949, 2024.
8. Y. Jiao, Y. Gu, T.-H. Chang and Z.-Q. Luo, "Decentralized Rank-Adaptive Matrix FactorizationPart II: Convergence Analysis," *IEEE Transactions on Signal Processing*, 2024.
9. K. Li, Y. Li, L. Cheng and Z.-Q. Luo, "Enhancing Multi-Stream Beamforming Through CQIs for 5G NR FDD Massive MIMO Communications: A Tuning-Free Scheme," *IEEE Transactions on Wireless Communications*, Vol. 23, No. 11, pp. 17508–17521, Nov. 2024.
10. S. Zhang, X. Ning, X. Zheng, Q. Shi, T.-H. Chang and Z.-Q. Luo, "A Physics-Based and Data-Driven Approach for Localized Statistical Channel Modeling," *IEEE Transactions on Wireless Communications*, Vol. 23, No. 6, pp. 5409–5424, June 2024.
11. K. Li, Y. Li, L. Cheng, Q. Shi and Z.-Q. Luo, "Downlink Channel Covariance Matrix Reconstruction for FDD Massive MIMO Systems With Limited Feedback," *IEEE Transactions on Signal Processing*, Vol. 72, pp. 1032–1048, 2024.
12. Q. Wu, B. Zheng, C. You, L. Zhu, K. Shen, X. Shao, W. Mei, B. Di, H. Zhang, E. Basar, L. Song, M. Di Renzo, Z.-Q. Luo, and R. Zhang, "Intelligent Surfaces Empowered Wireless Network: Recent Advances and the Road to 6G," *Proceedings of the IEEE*, 2024.
13. X. Zhao, S. Lu, Q. Shi and Z.-Q. Luo, "Rethinking WMMSE: Can Its Complexity Scale Linearly With the Number of BS Antennas?" *IEEE Transactions on Signal Processing*, Vol. 71, pp. 433–446, 2023.

14. Y.-B. Zhao and Z.-Q. Luo, "Dynamic Orthogonal Matching Pursuit for Sparse Data Reconstruction," *IEEE Open Journal of Signal Processing*, Vol. 4, pp. 242–256, 2023.

15. Z.-Q. Luo, X. Zheng, D. Lpez-Prez, Q. Yan, X. Chen, N. Wang, Q. Shi, T.-H. Chang, and A. Garcia-Rodriguez, "SRCON: A Data-Driven Network Performance Simulator for Real-World Wireless Networks," *IEEE Communications Magazine*, Vol. 61, No. 6, pp. 96–102, June 2023.

16. W. Pu, Y.-F. Liu and Z.-Q. Luo, "Efficient Estimation of Sensor Biases for the 3-D Asynchronous Multi-Sensor System," *IEEE Transactions on Signal Processing*, Vol. 71, pp. 2420–2433, 2023.

17. C. Chen, L. Shen, W. Liu and Z.-Q. Luo, "Efficient-Adam: Communication-Efficient Distributed Adam," *IEEE Transactions on Signal Processing*, Vol. 71, pp. 3257–3266, 2023.

18. M. Asgarian, G. Mirjalily and Z.-Q. Luo, "Embedding Multicast Service Function Chains in NFV-Enabled Networks," *IEEE Communications Letters*, Vol. 25, No. 4, pp. 1264–1268, April 2021.

19. Y. Yuan, G. Zheng, K.-K. Wong, B. Ottersten and Z.-Q. Luo, "Transfer Learning and Meta Learning Based Fast Downlink Beamforming Adaptation," *IEEE Transactions on Wireless Communications*, Nov. 2020.

20. F. Yin, L. Pan, T. Chen, S. Theodoridis, Z.-Q. Luo and A. M. Zoubir, "Linear Multiple Low-Rank Kernel Based Stationary Gaussian Processes Regression for Time Series," *IEEE Transactions on Signal Processing*, Vol. 68, pp. 5260–5275, 2020.

21. M. Hong, T.-H. Chang, X. Wang, M. Razaviyayn, S. Ma and Z.-Q. Luo, "A Block Successive Upper Bound Minimization Method of Multipliers for Linearly Constrained Convex Optimization," *Mathematics of Operations Research*, Vol. 68, pp. 2244–2255, June 2020.

22. Y. Huang, S. A. Vorobyov and Z.-Q. Luo, "Quadratic Matrix Inequality Approach to Robust Adaptive Beamforming for General-Rank Signal Model," *IEEE Transactions on Signal Processing*, Vol. 68, pp. 2244–2255, Jan. 2020.

23. Y. Yang, M. Pesavento, Z.-Q. Luo and B. Ottersten, "Inexact Block Coordinate Descent Algorithms for Nonsmooth Nonconvex Optimization," *IEEE Transactions on Signal Processing*, Vol. 68, pp. 947–961, Dec. 2019.

24. R. Sun, Z.-Q. Luo and Y. Ye, "On the Efficiency of Random Permutation for ADMM and Coordinate Descent," *Mathematics of Operations Research*, Vol. 45, pp. 233–271, Nov. 2019.

25. N. Moehle, X. Shen, Z.-Q. Luo and S. Boyd, "A Distributed Method for Optimal Capacity Reservation," *Journal of Optimization Theory and Applications*, Vol. 182, pp. 1130–1149, Sep. 2019.

26. Z. Lin, W. Pu and Z.-Q. Luo, "Minimax Design of Constant Modulus MIMO Waveforms for Active Sensing," *IEEE Signal Processing Letters*, Vol. 26, No. 10, pp. 1531–1535, Oct. 2019.

27. M. Razaviyayn, M. Hong, N. Reyhanian and Z.-Q. Luo, "A Linearly Convergent Doubly Stochastic GaussSeidel Algorithm for Solving Linear Equations and a Certain Class of Overparameterized Optimization Problems," *Mathematical Programming*, Vol. 176, pp. 465–496, July 2019.

28. W. Liu, R. Sun and Z.-Q. Luo, "Globally Optimal Joint Uplink Base Station Association and Beamforming," *IEEE Transactions on Communications*, May 2019.

29. A. Aubry, A. De Maio, A. Zappone, M. Razaviyayn and Z.-Q. Luo, "A New Sequential Optimization Procedure and Its Applications to Resource Allocation for Wireless Systems," *IEEE Transactions on Signal Processing*, Vol. 66, No. 24, pp. 6518–6533, Dec. 2018.

30. F. Shang, J. Cheng, Y. Liu, Z.-Q. Luo and Z. Lin, "Bilinear Factor Matrix Norm Minimization for Robust PCA: Algorithms and Applications," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 40, No. 9, pp. 2066–2080, Sep. 2018.

31. G. Mirjalily and Z.-Q. Luo, "Optimal Network Function Virtualization and Service Function Chaining: A Survey," *Chinese Journal of Electronics*, Vol. 27, No. 4, pp. 704–717, July 2018.

32. W. Liao, M. Hong, H. Farmanbar and Z.-Q. Luo, "A Distributed Semiasynchronous Algorithm for Network Traffic Engineering," *IEEE Transactions on Signal and Information Processing over Networks*, Vol. 4, No. 3, pp. 436–450, Sep. 2018.

33. W. Pu, Y. F. Liu, J. Yan, H. Liu and Z.-Q. Luo, "Optimal Estimation of Sensor Biases for Asynchronous Multi-Sensor Data Fusion," *Mathematical Programming*, Vol. 170, pp. 357–386, July 2018.

34. W. Pu, J. Xiao, T. Zhang and Z.-Q. Luo, "An Optimization Model for Electroencephalography-Assisted Binaural Beamforming," *The Journal of the Acoustical Society of America*, Vol. 143, No. 3, pp. 1744, Mar. 2018.

35. M. Hong and Z.-Q. Luo, "On the Linear Convergence of the Alternating Direction Method of Multipliers," *Mathematical Programming*, Vol. 162, pp. 165–199, Mar. 2017.

36. M. Hong and Z.-Q. Luo, "Iteration Complexity Analysis of Block Coordinate Descent Methods," *Mathematical Programming*, Vol. 163, pp. 85–114, May 2017.

37. M. Locatelli and Z.-Q. Luo, "On the Complexity of Optimal Power Allocation in a Multi-Tone Multiuser Communication System," *IEEE Transactions on Information Theory*, Vol. 63, pp. 6622–6627, Oct. 2017.

38. F. Shang, J. Cheng, Y. Liu, Z.-Q. Luo and Z. Lin, "Bilinear Factor Matrix Norm Minimization for Robust PCA: Algorithms and Applications," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2017.

39. N. Zhang, Z. Yao, Y. Liu, S. P. Boyd and Z.-Q. Luo, "Dynamic Resource Allocation for Energy Efficient Transmission in Digital Subscriber Lines," *IEEE Transactions on Signal Processing*, Vol. 65, pp. 4353–4366, Aug. 2017.

40. N. Zhang, Y. Liu, H. Farmanbar, T. Chang, M. Hong and Z.-Q. Luo, "Network Slicing for Service-Oriented Networks Under Resource Constraints," *IEEE Journal on Selected Areas in Communications*, Vol. 35, pp. 2512–2521, Nov. 2017.

41. Y.-B. Zhao, H. Jiang and Z.-Q. Luo, "Weak Stability of ℓ_1-Minimization Methods in Sparse Data Reconstruction," *IEEE Transactions on Signal and Information Processing over Networks*, Nov. 2017.

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112. Luo, Z.-Q. and Tseng, P., "Convergence Studies of Matrix Splitting Algorithms for the Linear Complementarity Problems," presented at the ORSA/TIMS conference, Nashville, Tennessee, May, 1991.

113. Luo, Z.-Q. and Tseng, P., "On the Hoffman's Error Bound for a Polyhedral Set," presented at the 14th International Symposium on Mathematical Programming, Amsterdam, The Netherlands, August, 1991.

114. Luo, Z.-Q. and Tsitsiklis, J., "On the Communication Complexity of Solving a Polynomial Equation," presented at *1990 International Symposium on Information Theory*, San Diego, California, January, 1990.

115. Luo, Z.-Q. and Tseng, P., "Convergence Studies of Coordinate Descent Algorithms for Convex Minimization Problems," presented at the ORSA/TIMS conference, Philadelphia, October, 1990.

116. Luo, Z.-Q. and Tsitsiklis, J., "Communication Complexity of Algebraic Computation," *1990 IEEE Symposium on Foundations of Computer Science*, pp. 758–765, October, 1990.

117. Luo, Z.-Q., "Communication Complexity of Computing a Collection of Rational Functions," presented on *International Conference on Computing and Information*, pp. 408–412, 1990, Niagara Falls, Ontario, Canada.

118. Jin, Q., Luo, Z.-Q. and Wong, K.M., "Optimum Signal Design in Time-Frequency Plane," *Proceedings of the International Symposium on Digital Signal Processing*, Beijing, P.R. China, October, 1990.

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C. Books and Special Issues

1. S. Cui, A. Hero, J. Moura and Z.-Q. Luo, *Big Data Over Networks*, Cambridge University Press, 400 pages, 2016.

2. Luo, Z.-Q, Pang, J.-S. and Ralph, D., *Mathematical Programs with Equilibrium Constraints*, Cambridge University Press, 400 pages, 1996.

3. Luo, Z.-Q. and Pang, J.-S. (Guest Editors), *Error Bounds and Their Applications in Mathematical Programming*, **Mathematical Programming**, Series B, 2000.

4. Chiang, M., Low, S., Luo, Z.-Q., Shroff, N. and Yu, W. (Guest Editors), Special issue of *IEEE Journal of Selected Areas of Communications* on 'Nonlinear Optimization of Communication Systems', 2006.

5. Luo, Z.-Q., Gastpar, M., Liu, J. and Swami, A., (Guest Editors), Special issue of *IEEE Signal Processing Magazine* on 'Distributed Signal Processing for Sensor Networks', 2006.

Students (2003 – present)

- Ph.D Students: Yushun Zhang, Ziniu Li, Juntao Wang, Dmitry Rybin, Yeqing Qiu, Yi Feng, Runhan Yang, Yuyang Zhao, Shangjie Zhou
- Graduated Ph.D Students:
  - Congliang Chen (Research Assistant Professor, Shenzhen Loop Area Institute, Shenzhen, PR China)
  - Ruoyu Sun (Associate Professor, The Chinese University of Hong Kong, Shenzhen, PR China)
  - Meisam Razaviyayn (Associate Professor, University of Southern California, USA)
  - Wenqiang Pu (Research Scientist, Shenzhen Research Institute of Big Data, PR China)
  - Jiawei Zhang (Assistant Professor, University of Wisconsin–Madison, USA)
  - Jiancong Xiao (Postdoc, University of Pennsylvania, USA)
  - Shuyi Ren (Postdoc, Linkping University, Sweden)
  - Congliang Chen (Research Assistant Professor, Shenzhen Loop Area Institute, Shenzhen, China)
  - Qingyan Meng (Research Assistant Professor, Pengcheng National Laboratory, China)
  - Liangqi Liu (Assistant Researcher, Qianyuan Laboratory, Hangzhou, China)
  - Tianjian Zhang (Senior Engineer, Huawei Technologies Co. Ltd, China)
  - Kai Li (Senior Engineer, Huawei Technologies Co. Ltd, China)
  - Hao Liang (Postdoc, King's College London, UK)
  - Jinjun Xiao (Senior Engineer, Starkey Hearing Technologies)
  - Alireza Razavi (Senior Member of Technical Staff, Aquantia)
  - Yao Huang (Senior Data Scientist, Intuit)
  - Mikalai Kiasiliou (Senior Engineer, Intel Corporation)
  - Maziar Sanjabi (Staff Engineer, Starkey Hearing Technologies)
  - Andy Tseng (Staff Engineer, Pindrop Security)
  - Wei-cheng Liao (Staff Engineer, Broadcom)
- Graduated Master Students: Ye Liu, Jingwei Mao, Boyuan Wang, Sha Lai, Ying Li, Liping Tang, Randy Platt, Ningyuan Wang, Chris Hamilton
- Postdoctoral fellows:
  - Mingyi Hong (Assoc. Prof., University of Minnesota, USA)
  - Zhiguo Wang (Assoc. Prof., Sichuan University, PR China)
  - Baldur Steingrimsson (Instructor, Oregon State University, USA)

INVITED PRESENTATIONS (2002 – present)

1. Keynote speaker, KAUST Research Conference on Mathematical and Data Sciences, Thuwal, Saudi Arabia, January 2026.
2. Keynote speaker, The 2nd Conference on Mathematics and Artificial Intelligence Sciences (Math & AI 2025), Dongguan, China.
3. Keynote speaker, CCF ChinaSoft 2025, Wuhan, China, November, 2025.
4. Invited speaker, the True Tech Day, MTS, Russia, June, 2025.
5. Keynote speaker, The 15th National Conference on Mathematical Optimization of the Mathematical Programming, Shanghai, China, May, 2025.
6. Keynote speaker, DO x ML - Seventh Conference on Discrete Optimization and Machine Learning, Kyoto, Japan, May, 2025.
7. Keynote speaker, IEEE PIMRC 2024, Valencia, Spain.
8. Keynote speaker, CSIAM-BDAI2024, Yinchuan, China.
9. Invited speaker, ICIAM 2023, Japan,Tokyo, August, 2023.
10. Invited speaker, The 14th International Conference on Numerical Optimization and Numerical Linear Algebra, Taiyuan, China, August 2023.
11. Keynote speaker, 18th International Symposium on Wireless Communication Systems (ISWCS2022), Hangzhou, China.
12. Plenary speaker, The 10th International Conference on Control, Automation and Information Science (ICCAIS 2021), Shanghai, China, October 2021
13. Plenary speaker, Chinese Congress on Artificial Intelligence, Chengdu, China, August 2021
14. Plenary speaker, 2021 PKU Workshop on Optimization Theory and Methods, Beijing, China, January 2021.
15. Plenary speaker, IEEE SPAWC 2020, Virtual Conference.
16. Plenary speaker, The 35th Youth Academic Annual Conference of Chinese Association of Automation, Zhanjiang, China, October 2020.
17. Plenary speaker, The Greater Bay Area Workshop on Computational Optimization, Department of Applied Mathematics, The Hong Kong Polytechnic University, January 2019.
18. Plenary speaker, Workshop on Data Driven Optimization Methods and Applications, Jilin University, Changchun, China, July 2019.
19. Keynote speaker, IEEE/CIC International Conference on Communications in China, Changchun, China, August 2019.

20. Invited speaker, The Third Greater Bay Area Precision Medicine Summit Forum, Shenzhen Peoples Hospital, August 2019.

21. Invited speaker, The 17th Annual Meeting of China Society for Industrial and Applied Mathematics, Foshan, China, September 2019.

22. Invited speaker, The 31st RAMP Symposium (RAMP2019), Keio University Yagami Campus, Tokyo, Japan, November 2019.

23. Keynote speaker, Workshop on Advances in Wireless Communications, Southeast University, Nanjing, China, May 4, 2018.

24. Plenary speaker, IEEE SAM, Sheffield, UK, July 2018.

25. Plenary speaker, CHIMA (China Hospital Information Management Association Annual Conference), Guizhou, China, July 2018.

26. Plenary speaker, Workshop on New Computing-Driven Opportunities for Optimization, Fujian, China, August 2018.

27. Colloquium speaker, Department of Mathematics, The Hong Kong University of Science and Technology, September 2018.

28. Keynote speaker, WCSP2018, Hangzhou, China, October 2018.

29. Plenary speaker, Huawei Wireless Algorithm Department, Shanghai, China, December 2018.

30. Invited speaker, Advances in Operations Research and Operations Management, City University of Hong Kong, January 2017.

31. Invited speaker, Conference on Nonconvex Statistical Learning, University of Southern California, May 2017.

32. Colloquium speaker, The Chinese University of Hong Kong, Shenzhen, August 2017.

33. Keynote speaker, IEEE ICCS 2016, Shenzhen, China.

34. Colloquium speaker, Department of Electrical and Computer Engineering, State University of North Carolina, November 2016.

35. Colloquium speaker, Department of Industrial Engineering, Northwestern University, May, 2016.

36. Plenary speaker, Huawei Wireless Algorithm Day, Shanghai, April, 2016.

37. Plenary speaker, ChinaCom, Chongqing, China, 2016.

38. Plenary speaker, IEEE ICCC, Shenzhen, China, 2015.

39. Colloquium speaker, Department of Electrical and Computer Engineering, University of Iowa, October, 2015.

40. Colloquium speaker, Department of Mathematics, McGill University, Montreal, Canada, August, 2014.

41. Plenary speaker, 2014 IEEE Communication Theory Workshop (CTW), Curacao.

42. Plenary speaker, 2013 MOPTA Conference, Lehigh University.

43. Invited speaker, 2013 Erice Workshop on Nonlinear Optimization, Erice, Italy.

44. Invited lecture, Collaborative Research Center "Highly Adaptive Energy-Efficient Computing", Technical University Dresden, Germany, 2013.

45. Plenary speaker, 2013 IEEE Signal Processing Advance for Wireless Communications (SPAWC) Workshop, Darmstad, Germany.

46. Invited speaker, 2013 summer school on the topic of Multi-Agent Networked Systems, Organizers: Tamer Basar, Angelia Nedich, Jong-Shi Pang, UIUC, 2013.

47. Overview/trend speaker, the IEEE China Summit and International Conference on Signal and Information Processing (ChinaSIP), Beijing, 2013.

48. Invited speaker, Department of Automation, University of Science and Technology of China, 2012.

49. Invited speaker, School of Telecommunication Engineering, Xidian University, Xi'an, China, 2012.

50. Colloquium speaker, Department of Electrical and Computer Engineering, Rutgers University, November, 2011.

51. Colloquium speaker, Department of Mathematics, KTH Royal Institute of Technology, October, 2011.

52. Semi-plenary speaker, *2011 IEEE CDC conference.*

53. Colloquium speaker, LIDS, MIT, November, 2010.

54. Semi-plenary speaker, 2010 International Conference on Continuous Optimization, Santiago, Chile.

55. Colloquium speaker, Department of Electrical Engineering, University of Iowa, October, 2009.

56. Invited speaker, 2009 Summer workshop on Optimization and Supply Chain Management, NCTU, Hsingchu, July 2009.

57. Colloquium speaker, Yuan-Tse University, Taiwan, August, 2009.

58. Invited speaker, 2009 International Conference on Numerical Linear Algebra and Nonlinear Optimization, Lijiang, China, August, 2009.

59. Invited speaker, 2009 International Conference on Linear Programming and Its Extensions, Haikou, China, May 2009.

60. Invited speaker, Huawei Technologies Inc., Shenzhen, China, May 2009.

61. Invited speaker, 2009 International Conference on Acoustic, Statistical Signal Processing, Taipei, March, 2009.

62. Invited speaker, 2009 Workshop on Convex Optimization and its Applications, University of Sharjah, the United Arab Emirates, April, 2009.

63. Invited speaker, The 4th Sino-Japanese Optimization Meeting (SJOM 2008), Tainan, August 2008.

64. Colloquium speaker, Department of Electrical Engineering, Princeton University, March, 2008.

65. Colloquium speaker, Department of Industrial and Enterprise Systems Engineering, UIUC, March 2008.

66. Colloquium speaker, Department of Industrial and Enterprise Systems Engineering, UIUC, October 2007.

67. Colloquium speaker, Department of Electrical and Computer Engineering, Syracuse University, October 2007.

68. Invited speaker, Department of Electrical and Computer Engineering, Texas A&M University, October 2007.

69. Invited speaker, Signals and Systems Distinguished Seminar Series, Department of Electrical Engineering, UCLA, October 2007.

70. Invited speaker, Distinguished Lecture Series, Department of Electrical Engineering, KTH, Sweden, December 2006.

71. Invited speaker, 2006 BIRS Workshop on Optimization and Engineering, Banff, November, Canada.

72. Plenary speaker, 2006 MOPTA Conference, Waterloo, Canada, July 2006.

73. Invited speaker, 2006 Annual Meeting of the Operations Research Society of China, Shenzhen, China, June 2006.

74. Invited speaker, Microsoft Research, Redmond, Washington, August 2006.

75. Invited speaker, Hughes Network Systems, Germantown, Maryland, March 2006.

76. Distinguished Lecturer, *IEEE Sensor Array and Multichannel Signal Processing Workshop*, Waltham, MA, June 2006.

77. Distinguished Lecturer, Program in Computation, Design and Optimization (CDO), MIT, March 2006.

78. Colloquium speaker, Laboratory for Information and Decision Systems, MIT, March 2006.

79. Invited speaker, The third Sino-Japan Optimization Meeting, Singapore, November 2005.

80. Colloquium speaker, Cornell University, School of Operations Research and Industrial Engineering, February 2005.

81. Invited speaker, Distinguished Lecture Series, Department of Electrical and Computer Engineering, University of Waterloo, May 2004.

82. Colloquium speaker, Department of Electrical and Computer Engineering, UCLA, November 2003.

83. Invited speaker, Information Systems Laboratory, Department of Electrical Engineering, Stanford University, October, 2003.

84. Invited speaker, Department of EECS, UC Berkeley, October 2003.

85. Semi-plenary speaker, International Symposium on Mathematical Programming, Copenhagen, Denmark, August 2003.

86. Tutorial, IEEE International Conference on Acoustics Speech and Signal Processing, Orlando, Florida, 2002.

87. Invited speaker, Department of Industrial and Systems Engineering, and School of Electrical and Computer Engineering, Georgia Institute of Technology, April 2002.

88. Colloquium speaker, the Johns Hopkins University, Department of Mathematical Sciences, February 2002.